Seat No.: 205 Enrolment No. 2015

# NATIONAL FORENSIC SCIENCES UNIVERSITY

**M.Sc. Cyber Security**
**Semester – II – July - 2024**

**Subject Code: CTMSCS SII P1** **Date: 10/07/2024**
**Subject Name: Network Security**
**Time: 2:30 PM to 5:30 PM** **Total Marks: 100**

**Instructions:**
1. **Write down each question on a separate page.**

| | | Marks |
|---|---|---|
| **Q.1** | **Attempt any three.** | |
| (a) | Explain OSI Model. | 08 |
| (b) | Explain following terms : | 08 |
| | 1) DNS 2) DHCP 3) Proxy 4) Application Server | |
| (c) | Explain following terms : | 08 |
| | 1) Threat 2) vulnerability 3) attack surface 4) attack vector | |
| (d) | Discuss application of following devices with its full form: | 08 |
| | 1) IDS 2) IPS 3) SIEM 4) EDR | |
| **Q.2** | **Attempt any three.** | |
| (a) | Write Short notes on Penetration testing life cycle. | 08 |
| (b) | Explain following Terms : 1) Dumpster Diving 2) War driving | 08 |
| (c) | What are the different password cracking techniques and solutions for hardening? | 08 |
| (d) | What do you mean by Secure code? Explain with one example. | 08 |
| **Q.3** | **Attempt any three.** | |
| (a) | Apply Polybius Square Cipher: 1) NFSU 2) BUS | 08 |
| (b) | Explain working of RSA algorithms. | 08 |
| (c) | Discuss working of AES algorithm. | 08 |
| (d) | Explain Elliptic Curve Cryptography. | 08 |
| **Q.4** | **Attempt any two.** | |
| (a) | Describe the four-way handshake process used in WPA2. | 07 |
| (b) | Explain 802.11 protocols. | 07 |
| (c) | You are tasked with securing a wireless network for a small business. Briefly outline the key steps you would take to implement a robust security solution, considering factors such as network size, user needs, and budget constraints. | 07 |
| **Q.5** | **Attempt any two.** | |
| (a) | Write short notes on OSCAR . | 07 |

(b) Discusses traffic acquisition Softwares and its importance for network forensic investigation. 07

(c) Write a short Netflow.. 07

**--- End of Paper---**

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# NATIONAL FORENSIC SCIENCES UNIVERSITY

**M.Sc. Cyber Security - Semester - II**

**End Semester Exam - July-2024**

**Subject Code: CTMSCS SII P2** **Date: 11/07/2024**

**Subject Name: Malware Analysis**

**Time: 2:30 PM - 5:30 PM** **Total Marks: 100**

55

**Instructions:**

1. **Write down each question on a separate page.**
2. **Attempt all questions.**
3. **Make suitable assumptions wherever necessary.**
4. **Figures to the right indicate full marks.**

| | | | Marks |
|---|---|---|---|
| **Q.1** | | **Attempt any three.** | |
| | **(a)** | 1. Explain function and property of obfuscated malware.<br>2. Define the function of PEiD and PEView tool in malware analysis. | **08** |
| | **(b)** | What are registers? Explain different types of registers with instruction example. | **08** |
| | **(c)** | Elaborate the Process Explorer and Process Monitor tools of sysinternal Utility. | **08** |
| | **(d)** | Explain following instructions with respect to X86:<br>1. mov eax, ebx<br>2. mov eax, 0x42<br>3. mov eax, [0x4037C4]<br>4. mov eax, [ebx]<br>5. mov eax, [ebx+esi*4]<br>6. cmp eax,5<br>7. mov ecx, [x]<br>8. xor eax, 2 | **08** |
| | | | |
| **Q.2** | | **Attempt any three.** | |
| | **(a)** | Explain types of breakpoints: Software execution, Hardware execution, Conditional, and memory breakpoints. | **08** |
| | **(b)** | Diagnose what would be the C program from following assembly snapshot.<br>00401006 mov [ebp+var_4], 0<br>0040100D mov [ebp+var_8], 1<br>00401014 mov eax, [ebp+var_4]<br>00401017 add eax, 0Bh<br>0040101A mov [ebp+var_4], eax<br>0040101D mov ecx, [ebp+var_4]<br>00401020 sub ecx, [ebp+var_8] | **08** |

| | | | |
|---|---|---|---|
| | | 00401023 mov [ebp+var_4], ecx<br>00401026 mov edx, [ebp+var_4]<br>00401029 sub edx, 1<br>0040102C mov [ebp+var_4], edx<br>0040102F mov eax, [ebp+var_8]<br>00401032 add eax, 1<br>00401035 mov [ebp+var_8], eax<br>00401038 mov eax, [ebp+var_4]<br>0040103B cdq<br>0040103C mov ecx, 3<br>00401041 idiv ecx<br>00401043 mov [ebp+var_8], edx | |
| | (c) | (a) What is the goal of Malware analysis?<br>(b) Describe the function of disassembler and debugger in malware analysis. | 08 |
| | (d) | Explain common executable DLLs of windows with function services. | 08 |
| | | | |
| Q.3 | | **Attempt any three.** | |
| | (a) | Discuss behaviour of hook injection while malware launching. | 08 |
| | (b) | Provide the practical explanation about setting up the malware analysis lab with required tool and technique with proper configurations. | 08 |
| | (c) | Differentiate between Static and Dynamic Analysis in detail. | 08 |
| | (d) | 1. Differentiate between User mode and Kernel mode.<br>2. Compare Virus vs Worm. | 08 |
| | | | |
| Q.4 | | **Attempt any two.** | |
| | (a) | What is Cyber-Warfare? Explain with the help of a real-world case study with respect to Malwares. | 07 |
| | (b) | Explain PE File Headers and Sections. | 07 |
| | (c) | Explain malware sandboxing. How various parameters can be identified from sandbox-based malware analysis? | 07 |
| | | | |
| Q.5 | | **Attempt any two.** | |
| | (a) | Explain function of FLOSS and Regshot tool in Malware Analysis. | 07 |
| | (b) | Explain Stack memory with example. | 07 |
| | (c) | Write a note on reverse engineering tool and its effectiveness in malware analysis. | 07 |

**--- End of Paper---**

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# NATIONAL FORENSIC SCIENCES UNIVERSITY

**M. Sc. Cyber Security**
**Semester – II – July - 2024**

**Subject Code: CTMSCS SII P4** **Date:** 15/07/2024
**Subject Name:** Incident Response & Digital Forensics
**Time:** 02:30 PM to 5:30 PM **Total Marks:** 100

**Instructions:**

1. **Write down each question on separate page.**
2. **Attempt all questions.**
3. **Make suitable assumptions wherever necessary.**
4. **Figures to the right indicate full marks.**

**Q.1** **Answer the following (Attempt Any Three)** **Marks**

(a) Explain the categories of Incident Response. **08**

(b) What are the signs of an incident that the IRM interns should look for? Provide examples. **08**

(c) i) Explain Incident Response need in the cyber security. **08**
ii) Discuss goals of Incident Response.

(d) Explain different data classification techniques to the team in an urgent meeting about client information security. **08**

**Q.2** **Answer the following (Attempt Any Three)**

(a) As the cybersecurity lead for a pharmaceutical company facing a ransomware attack compromising sensitive data, outline your incident prioritization strategy based on informational impact. Provide an example for each impact category. **08**

(b) Explain preparation, identification and eradication phases of Incident handling. **08**

(c) What are the various roles within an incident response team and their primary responsibilities? **08**

(d) How can an organization determine the cost of an incident? Provide an example. **08**

**Q.3** **Answer the following (Attempt Any Three)**

(a) Prepare an Incident handler checklist taking a case scenario. **08**

(b) How does virtualization affect the phases of incident handling? **08**

(c) What are the main sources of logs in Incident Response Management? Describe three log sources and how each helps in addressing security incidents within an organization's network. **08**

(d) Discuss the phases Incident Response Management: Containment, Eradication & Recovery. **08**

**Q.4** **Answer the following (Attempt Any Two)**

(a) Describe Registry Artefacts and their significance in building a case in Digital Forensics. Provide examples and explain their relevance in investigations. **07**

(b) What are write blockers, and how do imaging and cloning differ in digital forensics and data recovery contexts? **07**

(c) Explain file system and their characteristics: **07**

i) exFAT file system

ii) FAT32 file system

**Q.5** **Answer the following (Attempt Any Two)**

(a) What is role of Chain of Custody? Explain how integrity can maintain in Digital Forensics. **07**

(b) Explain Digital Forensics and its various branches. Additionally, discuss the importance of Locard's Principle of Exchange in the context of Digital Forensics. **07**

(c) Consider the following Sysmon Event Log Entry and answer the asked question: **07**


```
{
 "Event": {
  "Timestamp": "2023-05-20T13:45:55.000Z",
  "EventID": 1,
  "Provider": "Microsoft-Windows-Sysmon",
  "Channel": "Microsoft-Windows-Sysmon/Operational",
  "Computer": "SERVER-EXAMPLE",
  "EventData": {
   "Image": "C:\\Windows\\System32\\notepad.exe",
   "ParentImage": "C:\\Windows\\System32\\explorer.exe",
   "ParentCommandLine": "explorer.exe /factory,
  {BF7C3D9E-8E7A-4AA2-BD1E-6F12832D4AD6}",
   "TargetImage": "C:\\Program Files\\UnwantedProgram\\
    unwanted.exe",
   "TargetCommandLine": "unwanted.exe -silent -noUI",
   "GrantedAccess": "0x1400",
   "LogonId": "0x3e8",
   "UtcTime": "2023-05-20 13:45:54.500"
  }
 }
}
```


Explain the following event log parameters and their significance of Timestamp, Provider, Channel. and identify the potential threat through the provided event log.

**--- End of Paper---**

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**M. Sc. Cyber Security**
**Semester – II – July - 2024**

**Subject Code: CTMSCS SII P3** **Date:** 12/07/2024
**Subject Name:** Mobile Security
**Time:** 02:30 PM to 05:30 PM **Total Marks:** 100

**Instructions:**

1. **Write down each question on separate page.**
2. **Attempt all questions.**
3. **Make suitable assumptions wherever necessary.**
4. **Figures to the right indicate full marks.**

| | | Marks |
|---|---|---|
| **Q.1** | **Answer the following (Attempt Any Three)** | |
| (a) | You have discovered that an Android application has several unpatched vulnerabilities listed in the OWASP Top 10. Using dynamic analysis tools such as MobSF and Frida, explain how you would go about exploiting these vulnerabilities. Provide a detailed plan on how you would document the findings and propose fixes to the development team. | 08 |
| (b) | You have been hired by a financial institution to conduct a comprehensive security audit of their newly developed Android banking application. The application handles sensitive financial transactions and user data. Outline a detailed approach for performing both static and dynamic analysis of the application using tools like MobSF, Drozer. Explain how you would identify potential security flaws, assess their impact, and recommend remediation steps to ensure the application's security. | 08 |
| (c) | Discuss the steps involved in the Android boot process and their security implications. | 08 |
| (d) | Describe the importance of reverse engineering in Android Application Security and importance of secure source code review in Android application security | 08 |
| **Q.2** | **Answer the following (Attempt Any Three)** | |
| (a) | During a routine security audit, you suspect that an Android application is leaking sensitive user data over the network. Describe the steps you would take to perform traffic analysis using tools like Wireshark and HTTP Proxy Interception. Explain how you would identify and mitigate any data leaks found during your analysis. | 08 |
| (b) | How can Drozer be used for security auditing in Android applications? | 08 |

(c) Explain the architecture of Android. 08

(d) What is insecure login and how can it be exploited in Android applications? 08

**Q.3 Answer the following (Attempt Any Three)**

(a) What is the Android Run Time (ART) and its importance? Also discuss the purpose of ARM Translator in Android security? 08

(b) Discuss the common input validation issues in Android applications. 08

(c) What are client-side injection attacks, and how do they affect Android applications? 08

(d) Explain the concept of Android partitions and their significance in security. 08

**Q.4 Answer the following (Attempt Any Two)**

(a) Describe the role of ADB commands in Android application pen-testing by taking atleast 5 ADB Commands. 07

(b) Discuss the strategy for mobile application security pen-testing. 07

(c) Describe the process of traffic analysis for Android devices and its significance in security. 07

**Q.5 Answer the following (Attempt Any Two)**

(a) You are tasked with pen-testing a newly developed Android application. The app includes features like user authentication, data storage, and API communication. Describe the steps you would take to identify and exploit potential vulnerabilities in this application. Include the tools and techniques you would use. 07

(b) A company's proprietary Android app has been leaked online, and you are assigned to assess the potential security risks involved. Explain how you would use reverse engineering tools like (JD-GUI, Hex-Dump, etc..) to analyze the app's code and identify any sensitive information or vulnerabilities that might have been exposed. 07

(c) Describe the use of APKTool in reverse engineering Android applications in detail. 07

**--- End of Paper---**

# NATIONAL FORENSIC SCIENCES UNIVERSITY

## PRACTICAL EXAMINATION JULY 2024

**Subject Code:** CTMSCS SII L1 **Date:** 22/07/2024

**Subject Name:** Network Security Lab **Total Marks:** 50

**Time:** 30 Minutes

**Instructions:**

1. Attempt all questions.
2. Figures to the right indicate full marks. SET-1: Roll Number ending with 1, 5, 9, 1
   SET-2: Roll Number ending with 2, 6, 10
   SET-3: Roll Number ending with 3, 7
   SET-4: Roll Number ending with 4, 8

| SET | | Questions | Marks |
|---|---|---|---|
| SET-1 | Q.1 | Write a Python program that implements the ***RSA encryption algorithm***. Demonstrate its usage by encrypting, decrypting and digital signature a given message. | 25 |
| | Q.2 | Imagine a scenario where a distributed denial of service (DDoS) attack is targeting your organization's web servers. Outline the steps you would take to mitigate the impact and prevent future occurrences. | |
| SET-2 | Q.1 | Configure Quality of Service parameters in ***Cisco Packet Tracer*** to prioritize specific types of network traffic of NFSU website. Test and verify the effectiveness of QoS settings in managing network bandwidth. | |
| | Q.2 | In the event of a security incident, such as a data breach or malware infection, describe the steps you would take to contain the incident, investigate the root cause, and communicate effectively with stakeholders. | |
| SET-3 | Q.1 | Investigate any government website.to gather information about its ownership, hosting details, technologies used, and any potential security vulnerabilities and prepare the document with respect to PENTEST. Explain how to extract metadata, identify potential attack vectors, and assess the security posture of any government website.<br>Tools: WHOIS lookup, builtwith.com, Nmap, Shodan, Dredis, etc. | |
| | Q.2 | An employee's laptop has been lost or stolen. Describe the steps you would take to ensure that sensitive data on the device remains secure and cannot be accessed by unauthorized individuals. | |
| SET-4 | Q.1 | Task: Perform a vulnerability assessment on a NFSU website to identify open ports, running services, and potential weaknesses. Compile a report with identified vulnerabilities and remediation suggestions.<br>Tools: Nmap, Nessus, OpenVAS, Metasploit, etc.<br>Objectives: Discover network vulnerabilities, evaluate risk levels, and propose security improvements to harden the network. | |
| | Q.2 | Imagine an employee falls victim to a phishing attack and unknowingly installs malware on their workstation. Detail the measures you would take to detect and remove the malware, as well as prevent similar incidents in the future. | |

**END OF PAPER**